

Implement active-active replication with RDS Custom for Oracle: Part 1 – High Availability

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[Amazon Relational Database Service \(Amazon RDS\) Custom](#) is a managed database service for legacy, custom, and packaged applications that require access to the underlying OS and DB environment. With RDS Custom for Oracle, you can now access and customize your database server host and operating system, for example by applying special patches and changing the database software settings to support third-party applications that require privileged access. Through the time-saving benefits of a managed service, Amazon RDS Custom for Oracle allows you to focus on more important business-impacting, strategic activities. By automating backups and other operational tasks, you can rest easy knowing your data is safe and ready to be recovered if needed. Finally, Amazon RDS Custom cloud-based scalability can help your database infrastructure keep pace as their business grows. For more information about Amazon RDS Custom, refer to [Amazon RDS Custom for Oracle – New Control Capabilities in Database Environment](#).

Currently, you can achieve high availability (HA) on Amazon RDS Custom for Oracle by creating [read replicas](#), which you can also [promote](#) to standalone databases if the need arises.

In this post, we show you how to use [Oracle GoldenGate](#) (GoldenGate) to set up bi-directional replication on Amazon RDS Custom for Oracle. GoldenGate collects, replicates, and manages transactional data between databases. It is a log-based change data capture (CDC) and replication software package used with Oracle databases for online transaction processing (OLTP) systems. GoldenGate creates trail files that contain the most recent changed data from the source database and then pushes these files to the target database. We also use [Amazon Elastic File System](#) (Amazon EFS) along with GoldenGate to provide high availability (HA) and disaster recovery (DR) with near zero data loss in case of an [Availability Zone](#) (AZ) failure.

Amazon EFS is designed to provide 99.999999999% (11 9's) of durability over a given year. You can use Amazon EFS replication or [AWS Backup](#) to guard your Amazon EFS Single-AZ file system against the loss of an Availability Zone.

You can further scale the architecture by creating up to five read replicas of each RDS Custom instance, and you can also configure manual HA with Oracle Data Guard.

Amazon RDS Custom for Oracle is a bring your own license (BYOL) service. You should make sure you have GoldenGate license along with an Oracle EE license.

For more information about Amazon RDS Custom for Oracle and its features, refer to [Working with Amazon RDS Custom](#).

Solution overview

We have two RDS Custom for Oracle instances running in two different AZ's within the same AWS Region, which provides high availability. We can also create a cross-region solution for DR purposes that would have different design considerations and support different recovery objectives.

Our solution has the following components:

- Two RDS Custom for Oracle instances running in two different AZ's.
- Two Amazon EFS file systems attached to each RDS Custom for Oracle instance.
- Oracle GoldenGate binaries installed on each of the instances, and residing on an [Amazon Elastic Block Store](#) (Amazon EBS) volume.
- The GoldenGate shared files are stored on an Amazon EFS (EFS) volume on each RDS Custom for Oracle instance.

GoldenGate can also be configured on EBS without using EFS, but using the EFS filesystem for trailfiles provides the following benefits

- EFS provides the High Availability in the event of instance failure. By default, every EFS file system object (such as directory, file, and link) is redundantly stored across multiple AZs for file systems using Standard storage classes.
- EFS mount point can be mounted across several EC2 instances within the region.
- As EFS is Highly Available EFS provides near Zero RPO, EBS RPO is dependent on the restore time of the Snapshot.

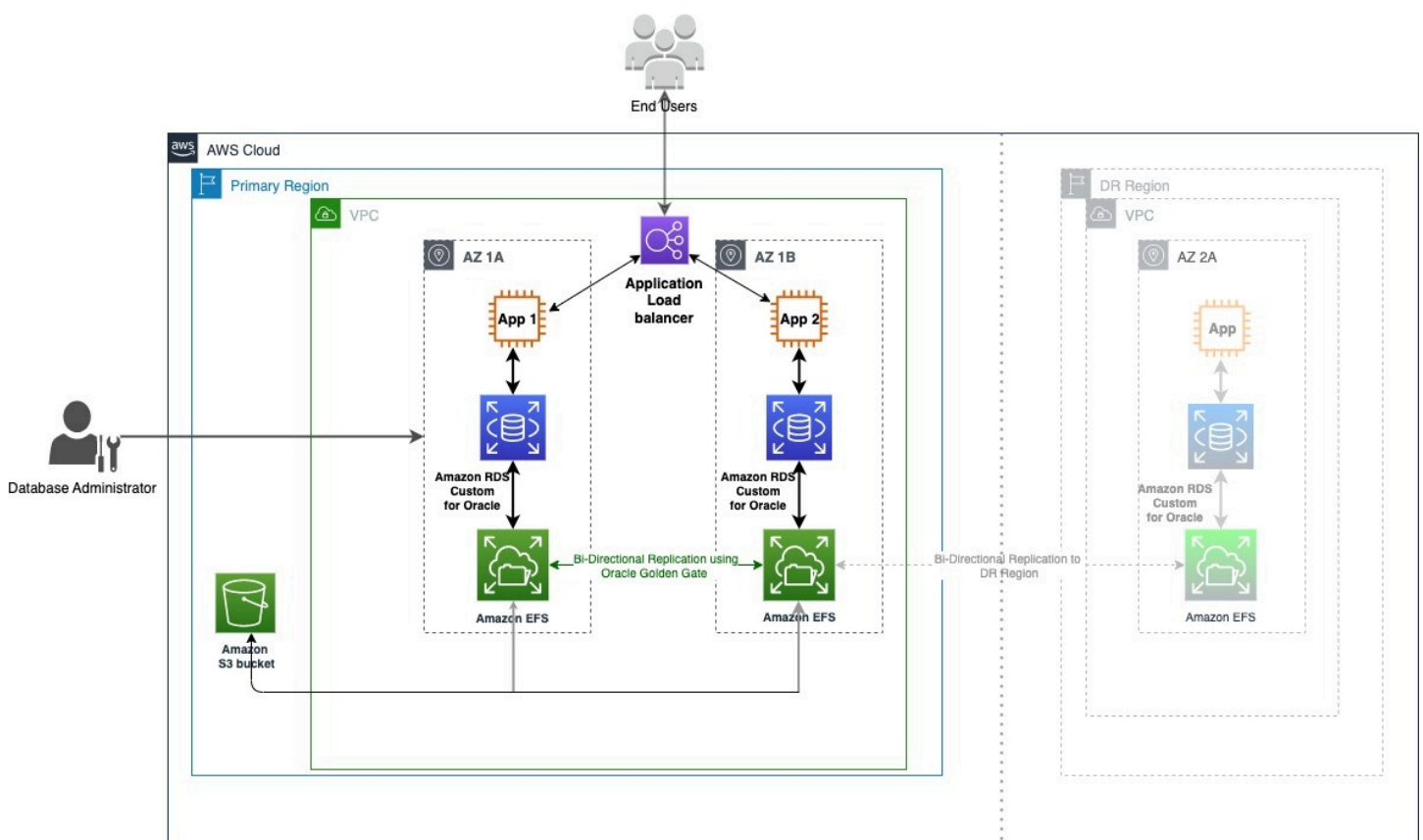
When deciding if you should be using file or block storage for your cloud application deployment, there are many factors to consider. The table below compares performance and storage characteristics for Amazon's highest performing file and block cloud storage offerings.

		File Amazon EFS	Block Amazon EBS
Performance	Per-operation latency	Low, consistent	Lowest, consistent
	Throughput scale	Multiple GBs per second	Single GB per second
Characteristics	Data Availability/Durability	Stored redundantly across multiple AZs	Stored redundantly in a single AZ
	Access	One to thousands of EC2 instances or on-premises servers, from multiple AZs, concurrently	Single EC2 instance in a single AZ

	Use Cases	Web serving and content management, enterprise applications, media and entertainment, home directories, database backups, developer tools, container storage, big data analytics
		Boot volumes, transactional and NoSQL databases, data warehousing & ETL

The following diagram illustrates the architecture for the RDS Custom for Oracle instance in a disaster recovery Region.

We have used in-Region Amazon RDS Custom for Oracle to implement HA, but we can also implement this as an HA/DR solution by creating the second Amazon RDS Custom instance in another Region .



This solution has the following advantages:

- Active-active RDS Custom for Oracle database instances support multiple writers in different geographies.
- Amazon EFS helps achieve a Recovery Point Objective (RPO) of near zero, meaning no data loss, because it holds the trail files even in case of AZ failure.
- You have an RPO of near zero by using Oracle bi-directional replication and Amazon EFS.
- With Amazon RDS Custom for Oracle, you can create up to five managed Oracle replicas of your RDS Custom for Oracle source DB instance in the same Region, which can provide read scaling or additional HA. You can also create your own manually configured (external) Oracle replicas in the same region or across regions which can provide HA/DR and read scaling. External replicas don't count toward your managed replica limit.

Considerations for an active-active configuration for Amazon RDS Custom for Oracle

The following considerations apply in an active-active configuration. For detailed explanations, refer to [Considerations for an Active-Active Configuration](#).

TRUNCATES

Bi-directional replication of `TRUNCATES` is not supported, but you can configure these operations to be replicated in one direction, while data is replicated in both directions. To replicate `TRUNCATES` (if supported by Oracle GoldenGate for the database) in an active-active configuration, the `TRUNCATES` must originate only from one database, and only from the same database each time.

Application design

When using active-active replication, the time zones must be the same on both systems so that timestamp-based conflict detection and resolution can operate.

Active-active replication is not recommended for use with commercially available packaged business applications, unless the application is designed to support it.

Keys

For accurate detection of conflicts, all records must have a unique, not-null identifier. If possible, create a primary key. If that is not possible, use a unique key or create a substitute key with a `KEYCOLS` option of the `MAP` and `TABLE` parameters. In the absence of a unique identifier, Oracle GoldenGate uses all the columns that are valid in a `WHERE` clause, but this will degrade performance if the table contains numerous columns.

Triggers and cascaded deletes

Triggers and `ON DELETE CASCADE` constraints generate DML operations that can be replicated by Oracle GoldenGate. To prevent the local DML from conflicting with the replicated DML from these operations, do the following:

- Modify triggers to ignore DML operations that are applied by GoldenGate Replicat. If the target is an Oracle database, Replicat handles triggers without any additional configuration when in [integrated mode](#). Parameter options are available for a nonintegrated Replicat for Oracle. For more information, see [12.1 Disabling Triggers and Referential Cascade Constraints on Target Tables](#).
- Disable `ON DELETE CASCADE` constraints and use a trigger on the parent table to perform the required record deletion on the child tables. Create it as a `BEFORE` trigger so that the child tables are deleted before the delete operation is performed on the parent table. This reverses the logical order of a cascaded delete but is necessary so that the operations are replicated in the correct order to prevent “table not found” errors on the target. For Oracle targets, if Replicat is in integrated mode, constraints are handled automatically without special configuration.

Database-generated values

Do not replicate database-generated sequential values, such as Oracle sequences, in a bi-directional configuration. The range of values must be different on each system, with no chance of overlap. For example, in a two-database environment, you can have one server generate even values, and the other odd. For an n -server environment, start each key at a different value and increment the values by the number of servers in the environment. This method may not be available to all types of applications or databases. If the application permits, you can add a location identifier to the value to enforce uniqueness.

Database configuration

One of the databases must be designated as the *trusted source*. This is the source database and its host system from which the other database is derived in the initial synchronization phase and in any subsequent resynchronizations that become necessary. Maintain frequent backups of the trusted source data.

Prevent data looping

In a bi-directional configuration, SQL changes that are replicated from one system to another must be prevented from being replicated back to the first system. Otherwise, it moves back and forth in an endless loop, as in this example:

1. A user application updates a row on system A.
2. GoldenGate Extract captures the change on system A and sends it to system B.
3. Replicat updates the row on system B.
4. GoldenGate Extract captures the change on system B and sends it back to system A.
5. The row is applied on system A (for the second time).
6. This loop continues endlessly.

To prevent data loopback, you may need to provide instructions that do the following:

- Prevent the capture of SQL operations that are generated by Replicat, but enable the capture of SQL operations that are generated by business applications if they contain objects that are specified in the Extract parameter file
- Identify local Replicat transactions, in order for the Extract process to ignore them

There are multiple ways to identify Replicat transaction in an Oracle environment. When Replicat is in classic or integrated mode, you use the following parameters:

- Use `DBOPTIONS` with the `SETTAG` option in the Replicat parameter file. Replicat tags the transactions being applied with the specified value, which identifies those transactions in the redo stream. The default `SETTAG` value is `00`. Valid values are a single TAG value consisting of hexadecimal digits. For more information about tags, see [Reference for Oracle GoldenGate for Windows and UNIX](#).
- Use the `TRANLOGOPTIONS` parameter with the `EXCLUDETAG` option in the Extract parameter file. The logmining server associated with that Extract excludes redo that is tagged with the `SETTAG` value.
- If you are excluding multiple tags, each must have a separate `TRANLOGOPTIONS EXCLUDETAG` statement specified.

- You can also use the transaction name or user ID of the Replicat user to identify Replicat transactions.

Manage conflicts

Uniform conflict-resolution procedures must be in place on all systems in an active-active configuration. Conflicts should be identified immediately and handled with as much automation as possible; however, different business applications present their own unique set of requirements in this area.

Because Oracle GoldenGate is an asynchronous solution, conflicts can occur when modifications are made to identical sets of data on separate systems at (or almost at) the same time.

To avoid conflicts, replication latency must be kept as low as possible. When conflicts are unavoidable, they must be identified immediately and resolved using the Oracle GoldenGate Conflict Detection and Resolution (CDR) feature, or through methods developed on your own. Custom methods can be integrated into Oracle GoldenGate processing through the SQLEXEC and user exit functionality. See [Configuring Conflict Detection and Resolution](#) for more information about using Oracle GoldenGate to handle conflicts.

We walk you through the following high-level steps:

1. Integrate an RDS Custom for Oracle instance with Amazon EFS.
2. Install GoldenGate on Amazon RDS Custom using Amazon EFS.
3. Set up a source database for use with GoldenGate on Amazon RDS Custom for Oracle.
4. Configure replication from the source RDS Custom for Oracle instance to the target RDS Custom for Oracle instance.
5. Configure replication from the target RDS Custom for Oracle instance to the source RDS Custom for Oracle instance.

Prerequisites

In order to follow along you must have the following:

- Two RDS Custom for Oracle instances running on [Amazon Elastic Compute Cloud](#) (Amazon EC2) with Oracle Linux, and the same Oracle Database software versions. The instances can be running in two different AZ's or different Regions. For instructions, see [Working with Amazon RDS Custom](#).
- The solution is applicable when the source database instance is on premises as long as you have appropriate network connectivity between the on-premises network and your [Amazon Virtual Private Cloud](#) (Amazon VPC).
- Connectivity between the two RDS Custom for Oracle instances, and the necessary ports opened for GoldenGate replication.

We walk you through the steps to install Oracle GoldenGate on Amazon RDS Custom for Oracle. We also cover the steps to configure bi-directional replication between two RDS Custom instances.

Follow the steps in [Creating and connecting to a DB instance for Amazon RDS Custom for Oracle](#) to create your two instances.

Implement Oracle GoldenGate on Amazon RDS Custom using Amazon EFS

Oracle GoldenGate collects, replicates, and manages transactional data between databases. We can use GoldenGate with Amazon RDS custom to do the following:

- Active-active database replication
- Disaster recovery
- Data protection
- In-Region and cross-Region replication
- Zero-downtime migration and upgrades

We can directly install GoldenGate on an RDS Custom for Oracle EC2 instance, but in case of failover, the RPO can be up to 5 minutes.

To achieve zero RPO, we integrate Amazon EFS with Amazon RDS Custom for Oracle, so that all the GoldenGate files that are required to replay the database transactions are stored on Amazon EFS. In a failover scenario, we can create a new read replica from the surviving database, mount the same Amazon EFS file system again, and start the bi-directional replication of GoldenGate to reestablish high availability.

You can also achieve the same resiliency if you decide to use one file system mounted across the two RDS Custom for Oracle DB instances. For this solution, we use two different Amazon EFS mount points, one for the source and one for the target, which helps make the management of GoldenGate easier and also makes it easier to troubleshoot.

Integrate an RDS Custom for Oracle instance with Amazon EFS

This integration has the following key benefits:

- Amazon EFS is used to store GoldenGate shared files.
- In failover scenarios, the same file system can be mounted on the new DB instance without risk of losing any data.
- Amazon EFS mounted on RDS Custom for Oracle DB instances as a landing zone for various Oracle files required for migration or data transfer. Since, Amazon EFS can be mounted across different AZ's and replicated cross regions enabling you to achieve faster migration.
- Using Amazon EFS as a landing zone also helps to save allocation of extra storage space on the instance to hold the files.
- Amazon EFS provides consistent performance to each compute instance. It's designed to be highly durable and highly available.
- The file systems can automatically scale from gigabytes to petabytes of data without needing to provision storage.

For integration instructions, refer to [Integrate Amazon RDS Custom for Oracle with Amazon EFS](#).

The following screenshot shows the Amazon RDS Custom integration with Amazon EFS on the source instance.

```
[ssm-user@ip-10-0-1-10 sudo su - rdsdb]
Last login: Wed Aug 10 17:20:50 UTC 2022 on pts/0
-bash-4.2$ df -h
Filesystem                                Size  Used Avail Use% Mounted on
devtmpfs                                 32G   0    32G   0% /dev
tmpfs                                    63G  28G   35G  45% /dev/shm
tmpfs                                    32G  644K   32G   1% /run
tmpfs                                    32G   0    32G   0% /sys/fs/cgroup
/dev/nvme0n1p1                           9.8G  8.4G   1.3G  88% /
/dev/mapper/dbdata01-lvdbdata01          197G  5.3G  190G   3% /rdsdbdata
/dev/nvme1n1                             25G  7.8G   16G  34% /rdsdbbin
tmpfs                                     6.3G   0    6.3G   0% /run/user/61001
tmpfs                                     6.3G   0    6.3G   0% /run/user/0
fs-10-0-1-10.efs.us-east-1.amazonaws.com:/ 8.0E  1.2G  8.0E   1% /ggate_efs_source
-bash-4.2$
```

The following screenshot shows the Amazon RDS Custom integration with Amazon EFS on the target instance.

```
[ssm-user@ip-10-0-1-10 sudo su - rdsdb]
Last login: Wed Aug 10 14:29:14 UTC 2022 on pts/0
-bash-4.2$
-bash-4.2$ df -h
Filesystem                                Size  Used Avail Use% Mounted on
devtmpfs                                 16G   0    16G   0% /dev
tmpfs                                    31G  14G   18G  46% /dev/shm
tmpfs                                    16G  644K   16G   1% /run
tmpfs                                    16G   0    16G   0% /sys/fs/cgroup
/dev/nvme0n1p1                           9.8G  7.9G   1.8G  82% /
/dev/mapper/dbdata01-lvdbdata01          197G  5.2G  190G   3% /rdsdbdata
/dev/nvme5n1                             25G  7.1G   17G  31% /rdsdbbin
tmpfs                                     3.1G   0    3.1G   0% /run/user/61001
tmpfs                                     3.1G   0    3.1G   0% /run/user/0
fs-10-0-1-10.efs.us-east-1.amazonaws.com:/ 8.0E  531M  8.0E   1% /ggate_efs_target
-bash-4.2$
```

Install Golden Gate on the source and target RDS Custom for Oracle instances

Follow the steps in this section for both instances.

Download Oracle GoldenGate compatible release version from [Obtaining the Oracle GoldenGate Distribution](#). To download the binaries for Oracle GoldenGate using WGET directly on to RDS Custom instance, follow the support doc WGET to download My Oracle Support (MOS) [Doc ID 980924.1](#)

Download the file `gg_download.sh` in user home and run the script to download the Oracle GoldenGate 19c software. This download require approximate 1 GB of additional space in the filesystem. Make sure the RDS Custom EC2 instance has internet access. For more information, refer to [Connect to the internet using an internet gateway](#).

The script prompts for the Oracle SSO user name and the password:

```
$ cd </Download_dir>
$ ./gg_download.sh
SSO User Name: <SSO ID>
Password: <SSO Password>
```

```
ls -lrt
-rw-rw-r-- 1 rdsdb rdsdb 556240981 Oct 23 2019 V983658-01.zip
-rwxrw-rw- 1 rdsdb rdsdb 2361 Jul 29 20:14 gg_download.sh
Unzip to extract the Golden Gate binaries.
```

Move the downloaded Zip file to the directory where you want to install the Golden binary files and then unzip the file as shown.

```
$ unzip V983658-01.zip
$ ls -lrt
-rw-rw-r-- 1 rdsdb rdsdb 1413 May 29 2019 OGG-19.1.0.0-README.txt
drwxr-xr-x 3 rdsdb rdsdb 6144 Oct 18 2019 fbo_ggs_Linux_x64_shiphome
-rw-rw-r-- 1 rdsdb rdsdb 332523 Oct 21 2019 OGG_WinUnix_Rel_Notes_19.1.0.0.4.pdf
-rw-rw-r-- 1 rdsdb rdsdb 556240981 Oct 23 2019 V983658-01.zip
-rwxrw-rw- 1 rdsdb rdsdb 2361 Jul 29 20:14 gg_download.sh
```

To install the GoldenGate software, we use the `./runInstaller` graphical user interface (GUI). Alternatively, silent mode installation allows us to configure necessary Oracle components without using a GUI. We can use the response file to provide all the required information for the installation, so no additional user input is required.

Take a backup of the original response file:

```
$ cd /<efs_location>/fbo_ggs_Linux_x64_shiphome/Disk1/response
$ cp oggcore.rsp oggcore.rsp.bk
$ ls -lrt
-rwxrwxr-x 1 rdsdb rdsdb 4439 Jan 23 2019 oggcore.rsp
-rwxrwxr-x 1 rdsdb rdsdb 4439 Aug 10 2022 oggcore.rsp.bk
```

To prepare the response file for GoldenGate installation, edit the following parameters and save the changes to the response file. Make sure to create required directories and provide the appropriate local filesystem location for the parameters.

```
$ cat oggcore.rsp
UNIX_GROUP_NAME=rdsdb
INSTALL_OPTION=ORA19c
SOFTWARE_LOCATION=/rdsdbbin/gg_home
START_MANAGER=true
MANAGER_PORT=7809
DATABASE_LOCATION=/rdsdbbin/oracle
INVENTORY_LOCATION=/rdsdbbin/gg_home/oraInventory
```

Now you can complete GoldenGate silent installation by running the following script. We also include the output of the silent installation.

```
$ cd /<efs_location>/fbo_ggs_Linux_x64_shiphome/Disk1
$ ./runInstaller -silent -showProgress -waitforcompletion -responseFile /<efs_location>/

Starting Oracle Universal Installer...
Checking Temp space: must be greater than 120 MB. Actual 1865 MB Passed
Checking swap space: must be greater than 150 MB. Actual 16383 MB Passed
You can find the log of this install session at
/rdsdbbin/gg_home/oraInventory/logs/installActions2022-08-10_07-15-11PM.log

Copy files successful.
Link binaries in progress.
.....Link binaries successful.
Setup files in progress.
..... 85% Done.
.....
Setup files successful.
Setup Inventory in progress.
Setup Inventory successful.
```

Run the following command as root user.

```
root@ip]# /rdsdbbin/gg_home/oraInventory/orainstRoot.sh
Changing permissions of /rdsdbbin/gg_home/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.
Changing groupname of /rdsdbbin/gg_home/oraInventory to rdsdb.
The execution of the script is complete.
```

Validate the golden gate installation by running the `ggsci` command.

```
$ cd /rdsdbbin/gg_home
$ ./ggsci

$ GGSCI (ip-address) > info all
Program      Status      Group Lag at Chkpt Time Since Chkpt
MANAGER RUNNING
```

After the GoldenGate installation we need to create required subdirectories through the [GoldenGate Software Command Interface \(GGSCI\)](#) using [create subdirs](#) command. Since we are using EFS, these subdirectories must be

created on Amazon EFS manually and then create symbolic links from GoldenGate home (also known as *soft links*) that point to Amazon EFS.

There is a two-step process to create the required 15 symbolic links.

- **Step 1:**

Create the subdirectories on Amazon EFS. The following commands shows how to create the subdirectories. You should run these commands using the RDS Custom for Oracle operating system user `rdsdb`. We assume the default owners and paths, make sure to adjust as appropriate for your Custom Oracle home. This operating system user is not the root user and doesn't require root privileges. For more information about required privileges, see [Operating System Privileges](#). In the following screenshot we created a directory `gg_data` under the EFS location and then created the directories under `gg_data`.

```
$ mkdir /<efs_location>/gg_data/BR
$ mkdir /<efs_location>/gg_data/br
$ mkdir /<efs_location>/gg_data/dirchk
$ mkdir /<efs_location>/gg_data/dirrpt
$ mkdir /<efs_location>/gg_data/dirpcs
$ mkdir /<efs_location>/gg_data/dirdat
$ mkdir /<efs_location>/gg_data/dirbdb
$ mkdir /<efs_location>/gg_data/dirprm
$ mkdir /<efs_location>/gg_data/dirwlt
$ mkdir /<efs_location>/gg_data/dircrd
$ mkdir /<efs_location>/gg_data/dirsq1
$ mkdir /<efs_location>/gg_data/dirdef
$ mkdir /<efs_location>/gg_data/dirtmp
$ mkdir /<efs_location>/gg_data/dirdmp
$ mkdir /<efs_location>/gg_data/dirdsc
```

- **Step 2:**

1. Create the symbolic links from the GoldenGate home directory.
2. Create a soft link for each with the same name pointing to the Amazon EFS location where we want the trail files to get generated. Make sure these directories are deleted from default GoldenGate home before running these commands. For example:

```
$ ln -s /<efs_location>/gg_data/BR BR
$ ln -s /<efs_location>/gg_data/br br
$ ln -s /<efs_location>/gg_data/dirchk dirchk
$ ln -s /<efs_location>/gg_data/dirrpt dirrpt
$ ln -s /<efs_location>/gg_data/dirpcs dirpcs
$ ln -s /<efs_location>/gg_data/dirdat dirdat
$ ln -s /<efs_location>/gg_data/dirbdb dirbdb
$ ln -s /<efs_location>/gg_data/dirprm dirprm
$ ln -s /<efs_location>/gg_data/dirwlt dirwlt
$ ln -s /<efs_location>/gg_data/dircrd dircrd
```

```
$ ln -s /<efs_location>/gg_data/dirsq1 dirsq1
$ ln -s /<efs_location>/gg_data/dirdef dirdef
$ ln -s /<efs_location>/gg_data/dirtmp dirtmp
$ ln -s /<efs_location>/gg_data/dirdmp dirdmp
$ ln -s /<efs_location>/gg_data/dirdsc dirdsc
```

The following screenshot shows how the soft link looks like from the Oracle GoldenGate home directory to the Amazon EFS file system.

```
lrwxrwxrwx 1 rdsdb rdsdb      28 Aug 10 20:05 BR -> /ggate_efs_target/gg_data/BR
lrwxrwxrwx 1 rdsdb rdsdb      28 Aug 10 20:05 br -> /ggate_efs_target/gg_data/br
lrwxrwxrwx 1 rdsdb rdsdb      32 Aug 10 20:05 dirbdb -> /ggate_efs_target/gg_data/dirbdb
lrwxrwxrwx 1 rdsdb rdsdb      32 Aug 10 20:05 dirdsc -> /ggate_efs_target/gg_data/dirdsc
-rw-r----- 1 rdsdb rdsdb    15399 Aug 11 22:01 ggerr.log
lrwxrwxrwx 1 rdsdb rdsdb      32 Aug 11 22:37 dirchk -> /ggate_efs_target/gg_data/dirchk
lrwxrwxrwx 1 rdsdb rdsdb      32 Aug 11 22:37 dirrpt -> /ggate_efs_target/gg_data/dirrpt
lrwxrwxrwx 1 rdsdb rdsdb      32 Aug 11 22:37 dirpcs -> /ggate_efs_target/gg_data/dirpcs
lrwxrwxrwx 1 rdsdb rdsdb      32 Aug 11 22:37 dirdat -> /ggate_efs_target/gg_data/dirdat
lrwxrwxrwx 1 rdsdb rdsdb      32 Aug 11 22:37 dirprm -> /ggate_efs_target/gg_data/dirprm
lrwxrwxrwx 1 rdsdb rdsdb      32 Aug 11 22:37 dirwlt -> /ggate_efs_target/gg_data/dirwlt
lrwxrwxrwx 1 rdsdb rdsdb      32 Aug 11 22:37 dircrd -> /ggate_efs_target/gg_data/dircrd
lrwxrwxrwx 1 rdsdb rdsdb      32 Aug 11 22:37 dirsqr -> /ggate_efs_target/gg_data/dirsqr
lrwxrwxrwx 1 rdsdb rdsdb      32 Aug 11 22:37 dirdef -> /ggate_efs_target/gg_data/dirdef
lrwxrwxrwx 1 rdsdb rdsdb      32 Aug 11 22:37 dirtmp -> /ggate_efs_target/gg_data/dirtmp
lrwxrwxrwx 1 rdsdb rdsdb      32 Aug 11 22:37 dirdmp -> /ggate_efs_target/gg_data/dirdmp
```

The higher-level gg directory can't be symbolically linked. The symbolic links must only be created for the dir* directories shown in the preceding screenshot. There are files specific to the GoldenGate installation that must not be shared across GoldenGate environments, specifically GoldenGate error logs, libraries, and executables.

Symbolic links are supported for GoldenGate subdirectories. Oracle has published a [My Oracle Support \(MOS\)](#) document validating this configuration. The MOS Doc ID is [1366597.1](#).

Perform the GoldenGate binary configuration steps on second instance following the same steps.

Set up a source database for use with GoldenGate on Amazon RDS Custom for Oracle

Complete the following prerequisite configurations for Oracle Database in order to set up GoldenGate on the source and target database. The following are the database-related parameters that you must configure before setting up GoldenGate.

```
SQL> ALTER SYSTEM SET ENABLE_GOLDENGATE_REPLICATION=TRUE SCOPE=BOTH;
SQL> ALTER DATABASE FORCE LOGGING;
SQL> ALTER DATABASE ADD SUPPLEMENTAL LOG DATA;
SQL> select LOG_MODE from v$database;
```

```
LOG_MODE
-----
ARCHIVELOG
```

```
SQL> select SUPPLEMENTAL_LOG_DATA_MIN, FORCE_LOGGING from v$database;
```

```
SUPPLEMENTAL_LOG_DATA_MIN  FORCE_LOGGING
```

```
-----
```

```
YES      YES
```

```
SQL> show parameter ENABLE_GOLDENGATE_REPLICATION;
```

```
NAME                                TYPE      VALUE
```

You must also create the administrator and user or schema owners on both the source and target database and grant required privileges. The following code [grants the necessary account privileges](#) to the GoldenGate user (for this post, `gguser`) for the RDS Custom for Oracle source and target environments. The best practice is to grant at least access for the GoldenGate user for the RDS Custom for Oracle database. For more information, visit [Granting the Appropriate User Privileges](#).

```
SQL> create user gguser identified by gguser default tablespace users quota unlimited;
SQL> grant create session,connect,resource,alter system to gguser;
SQL> exec dbms_goldengate_auth.grant_admin_privilege (grantee=>'gguser', privilege=>'all');
SQL> exec dbms_goldengate_auth.grant_admin_privilege (grantee=>'gguser', privilege=>'all');
SQL> GRANT CREATE SESSION, ALTER SESSION TO gguser;
SQL> GRANT RESOURCE TO gguser;
SQL> GRANT SELECT ANY DICTIONARY TO gguser;
SQL> GRANT FLASHBACK ANY TABLE TO gguser;
SQL> GRANT SELECT ANY TABLE TO gguser;
SQL> GRANT SELECT_CATALOG_ROLE TO gguser WITH ADMIN OPTION;
SQL> GRANT EXECUTE ON DBMS_FLASHBACK TO gguser;
SQL> GRANT SELECT ON SYS.V_$DATABASE TO gguser;
SQL> GRANT ALTER ANY TABLE TO gguser;
SQL> GRANT CREATE SESSION TO gguser;
SQL> GRANT ALTER SESSION TO gguser;
SQL> GRANT CREATE CLUSTER TO gguser;
SQL> GRANT CREATE INDEXTYPE TO gguser;
SQL> GRANT CREATE OPERATOR TO gguser;
```

Add the following entries to `$ORACLE_HOME/network/admin/tnsnames.ora` in the Oracle home to be used by the [EXTRACT](#) process. For more information on the `tnsnames.ora` file, see [Local Naming Parameters in the tnsnames.ora File](#).

```
RDSSOURCE=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=TCP)(HOST=<RDS for C
RDSTARGET=(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=TCP)(HOST=<RDS for C
```

Set up the manager process on the source database:

```
$ ./ggsci
$ GGSCI (ip-gg-source) > stop mgr
$ GGSCI (ip-gg-source) > edit param mgr

PORT 7810
DynamicPortList 7809-7899
PurgeOldExtracts ./dirdat/, UseCheckPoints, MinKeepHours 2
Autostart Extract E*
Autostart Extract P*
Autostart Replicat R*
AUTORESTART Extract * , WaitMinutes 1, Retries 3
AUTORESTART Replicat * , WaitMinutes 1, Retries 3

$ GGSCI (ip-gg-source) > start mgr
$ GGSCI (ip-gg-source) > info mgr
Manager is running (IP port TCP:ip-gg-source.7809, Process ID 2200).
```

Set up the manager process on the target database:

```
$ ./ggsci
$ GGSCI (ip-gg-source) > stop mgr
$ GGSCI (ip-gg-source) > edit param mgr

PORT 7809
DynamicPortList 7809-7899
PurgeOldExtracts ./dirdat/, UseCheckPoints, MinKeepHours 2
Autostart Extract E*
Autostart Extract P*
Autostart Replicat R*
AUTORESTART Extract * , WaitMinutes 1, Retries 3
AUTORESTART Replicat * , WaitMinutes 1, Retries 3

$ GGSCI (ip-gg-target) > start mgr
$ GGSCI (ip-gg-target) > info mgr
```

Configure a [GoldenGate credential store](#) to maintain encrypted database passwords and user IDs.

Use the following code for the source database:

```
$ ./ggsci
$ GGSCI (ip-gg-source) > ADD CREDENTIALSTORE
Credential store created.
$ GGSCI (ip-gg-source) > ALTER CREDENTIALSTORE ADD USER gguser@RDSCUSTOM_source pass
```

```
Credential store altered.
$ GGSCI (ip-gg-source) > dblogin USERIDALIAS rdscustom_source
Successfully logged into database.
$ GGSCI (ip-gg-source as gguser@RDSTEST) 5> INFO CREDENTIALSTORE
Reading from credential store:
Default domain: OracleGoldenGate
Alias: rdscustom_source
Userid: gguser@RDSCUSTOM_source

$ GGSCI (ip-gg-source as gguser@RDSTEST) 6> info all
Program Status Group Lag at Chkpt Time Since Chkpt
MANAGER RUNNING
```

Use the following code for the target:

```
$ ./ggsci
$ GGSCI (ip-gg-target) > ADD CREDENTIALSTORE
Credential store created.
$ GGSCI (ip-gg-target) > ALTER CREDENTIALSTORE ADD USER gguser@RDSCUSTOM_TARGET pass
Credential store altered.
$ GGSCI (ip-gg-target) > dblogin USERIDALIAS rdscustom_target
Successfully logged into database.
$ GGSCI (ip-gg-target as gguser@ORCL) > INFO CREDENTIALSTORE
Reading from credential store:
Default domain: OracleGoldenGate
Alias: rdscustom_target
Userid: gguser@RDSCUSTOM_TARGET

$ GGSCI (ip-gg-target as gguser@ORCL) > info all
Program Status Group Lag at Chkpt Time Since Chkpt
MANAGER RUNNING
```

Create an [Oracle GoldenGate checkpoint table](#) on the source and target by running the following code on each environment:

```
$ ./ggsci
$ GGSCI (ip-source as gguser@RDSTEST) > add checkpointtable gguser.chkpt
Successfully created checkpoint table gguser.chkpt.
$ GGSCI > info checkpointtable gguser.chkpt
```

Set up the demo schema and sample table (**Heartbeat**) by running the following SQL*Plus commands on each environment.

```
SQL>create user demo identified by demo default tablespace users quota unlimited on
SQL>grant create session,connect,resource,alter system to demo;
SQL>create table demo.heartbeat ( id number(10),heartbeat_date date, AZ varchar2(5)
SQL>create table demo.heartbeat ( id number(10),heartbeat_date date, AZ varchar2(5)
SQL>ALTER TABLE demo.heartbeat ADD ( CONSTRAINT id_pk source KEY (id)
```

The GoldenGate [ADD SCHEMATRANDATA](#) command acts on all current and future tables in a given schema to automatically enable schema-level supplemental logging. [ADD TRANDATA](#) for Oracle enables the unconditional logging of the primary key and the conditional supplemental logging of all unique keys and foreign keys of the specified table. If possible, use the ADD SCHEMATRANDATA command rather than the ADD TRANDATA command. The ADD SCHEMATRANDATA command ensures replication continuity should DML occur on an object for which DDL has just been performed

The following excerpt is for the source, using ADD SCHEMATRANDATA:

```
$ ./ggsci
$ GGSCI (ip-source) > dblogin USERIDALIAS rdscustom_source
Successfully logged into database.
$ GGSCI (ip-source as gguser@RDSTEST) > ADD SCHEMATRANDATA demo
$ GGSCI (ip-source as gguser@RDSTEST) > ADD TRANDATA demo.heartbeat ALLCOLS
$ GGSCI (ip-source as gguser@RDSTEST) > info SCHEMATRANDATA demo
$ GGSCI (ip-source as gguser@RDSTEST) > info trandata demo.heartbeat
Oracle Goldengate support native capture on table DEMO.HEARTBEAT.
Oracle Goldengate marked following column as key columns on table DEMO.HEARTBEAT: ID
Logging of supplemental redo log data is enabled for table DEMO.HEARTBEAT.
All columns supplementally logged for table DEMO.HEARTBEAT.
Prepared CSN for table DEMO.HEARTBEAT: 3651244
```

The following excerpt is for the target, using ADD SCHEMATRANDATA:

```
$ ./ggsci
$ GGSCI (ip-target) > dblogin USERIDALIAS rdscustom_target
Successfully logged into database.
$ GGSCI (ip-target as gguser@ORCL) > ADD SCHEMATRANDATA demo
$ GGSCI (ip-target as gguser@ORCL) > ADD TRANDATA demo.heartbeat ALLCOLS
$ GGSCI (ip-target as gguser@ORCL) > info SCHEMATRANDATA demo
$ GGSCI (ip-target as gguser@ORCL) > info trandata demo.heartbeat
Oracle Goldengate support native capture on table DEMO.HEARTBEAT.
Oracle Goldengate marked following column as key columns on table DEMO.HEARTBEAT: ID
Logging of supplemental redo log data is enabled for table DEMO.HEARTBEAT.
All columns supplementally logged for table DEMO.HEARTBEAT.
```

Validate the log group created in source and target database using following SQL*Plus commands:

```
SQL> select * from dba_log_groups where owner='DEMO';
OWNER LOG_GROUP_NAME TABLE_NAME LOG_GROUP_TYPE ALWAYS GENERATED
DEMO GGS_27751 HEARTBEAT USER LOG GROUP ALWAYS USER NAME
DEMO SYS_C006037 HEARTBEAT PRIMARY KEY LOGGING ALWAYS GENERATED NAME
DEMO SYS_C006038 HEARTBEAT UNIQUE KEY LOGGING CONDITIONAL GENERATED NAME
DEMO SYS_C006039 HEARTBEAT FOREIGN KEY LOGGING CONDITIONAL GENERATED NAME
```

Lastly, edit the **GLOBALS** parameter file on both the source and target GoldenGate home directory from the operating system. The **GLOBALS** file stores parameters that relate to the Oracle GoldenGate instance as a whole. The parameters in the **GLOBALS** file apply to all processes in the Oracle GoldenGate instance, but can be overridden by specific process parameters. From the Oracle GoldenGate installation location, run GGSCI and enter the following command, or open a file in a text editor.

```
$. /ggsci
$ GGSCI > EDIT PARAMS ./GLOBALS

GGSCHEMA gguser
CHECKPOINTTABLE gguser.chkpt
ALLOWOUTPUTDIR /<efs_filesystem>/gg_data/dirdat
```

Configure Replication from the source to target instance

To configure the source Extract group, use the EDIT PARAMS command to create a parameter file for the ECUST_S group. Include the following parameters plus any others that apply to your database environment. For possible additional required parameters, see the Oracle GoldenGate installation and setup guide for your database.

```
$ ./ggsci
$ GGSCI > edit params ECUST_S

EXTRACT ECUST_S
USERIDALIAS rdscustom_source
TranlogOptions IntegratedParams (max_sga_size 256)
TRANLOGOPTIONS EXCLUDETAG 00
ddl include all
LOGALLSUPCOLS
UPDATERECORDFORMAT COMPACT
EXTTRAIL ./dirdat/es
TABLE demo.*;
```

Run the following command on GGSCI prompt.

```
$GGSCI (ip-source) > edit params ECUST_S
$GGSCI (ip-source) > ADD EXTRACT ECUST_S, INTEGRATED TRANLOG, BEGIN NOW
EXTRACT (Integrated) added.
$GGSCI (ip-source) > ADD EXTTRAIL ./dirdat/es, EXTRACT ECUST_S
EXTTRAIL added.
$GGSCI (ip-source) > dblogin useridalias rdscustom_source
Successfully logged into database.

$GGSCI (ip-source as gguser@RDSTEST) > register extract ECUST_S database
2022-08-11 21:44:38 INFO OGG-02003 Extract ECUST_S successfully registered with data
```

To configure the data pump extract process, use the following GGSCI commands:

```
$ ./ggsci
$GGSCI (ip-source as gguser@RDSTEST) > edit param PCUST_S
Extract PCUST_S
UserIdAlias rdscustom_source
rmthost 10.0.156.210, mgrport 7809
rmtrail /<remote_efs_location>/es
Table demo.*;
```

Run the following command on GGSCI prompt.

```
$GGSCI (ip-source as gguser@RDSTEST) > add extract PCUST_S, EXTTRAILSOURCE ./dirdat/
EXTRACT added.
$GGSCI (ip-source as gguser@RDSTEST) > add rmtrail /<remote_efs_location>/es, extra
RMTRAIL added.
```

To configure the Replicat group, use the following GGSCI commands:

```
$ ./ggsci
$GGSCI > edit params RCUST_T

Replicat RCUST_T
UserIdAlias rdscustom_source
DBOPTIONS INTEGRATEDPARAMS(parallelism 4)
DBOPTIONS SETTAG 00
AssumeTargetDefs
```

```
DiscardFile ./dirrpt/rcust_t.dsc, Purge
Map demo.*, target demo.*;
```

Run the following command on GGSCI prompt.

```
--Use the ADD REPLICAT command to create a Replicat group
$GGSCI > ADD REPLICAT RCUST_T, INTEGRATED, EXTTRAIL ./dirdat/et
```

Configure Replication from the target to source instance

To configure the target Extract group, use the EDIT PARAMS command to create a parameter file for the ECUST_T group. Include the following parameters plus any others that apply to your database environment. For possible additional required parameters, see the Oracle GoldenGate installation and setup guide for your database.

```
$ ./ggsci
$GGSCI > edit params ECUST_T

EXTRACT ECUST_T
USERIDALIAS rdscustom_target
TranlogOptions IntegratedParams (max_sga_size 256)
TRANLOGOPTIONS EXCLUDETAG 00
ddl include all
LOGALLSUPCOLS
UPDATERECORDFORMAT COMPACT
EXTTRAIL ./dirdat/et
TABLE demo.*;
```

Run the following command on GGSCI prompt.

```
--Use the ADD EXTRACT command to create a source Extract group
$GGSCI > ADD EXTRACT ECUST_T, INTEGRATED TRANLOG, BEGIN NOW

--Use the ADD EXTTRAIL command to add a local trail.
$GGSCI > ADD EXTTRAIL ./dirdat/et, EXTRACT ECUST_T
$GGSCI > dblogin USERIDALIAS rdscustom_target
$GGSCI > register extract ECUST_T database
2022-08-12 18:49:26 INFO OGG-02003 Extract ECUST_T successfully registered with data
```

To configure the data pump extract process on the target, use the **EDIT PARAMS** GGSCI command to create a parameter file for the **PCUST_T** group:

```
$ ./ggsci
$GGSCI > edit param PCUST_T

Extract PCUST_T
UserIdAlias rdscustom_target
rmthost localhost, mgrport 7809
rmtrail /<remote_efs_location>/gg_data/dirdat/et
Table demo.*;
```

Run the following command on GGSCI prompt

```
$GGSCI > GGSCI (ip-target as gguser@ORCL) > edit param PCUST_T
$GGSCI > GGSCI (ip-target as gguser@ORCL) 1> add extract PCUST_T, EXTTRAILSOURCE ./c
EXTRACT added.
$GGSCI > GGSCI (ip-target as gguser@ORCL) 1> add rmtrail /<remote_efs_location>/gg_
RMTTRAIL added.
```

To configure the Replicat group, use the **EDIT PARAMS** command to create a parameter file for the **RCUST_S** group. Include the following parameters plus any others that apply to your database environment. For possible additional required parameters, see the Oracle GoldenGate installation and setup guide for your database.

```
$ ./ggsci
$GGSCI > edit params RCUST_S

Replicat RCUST_S
UserIdAlias rdscustom_target
DBOPTIONS INTEGRATEDPARAMS(parallelism 4)
DBOPTIONS SETTAG 00
AssumeTargetDefs
DiscardFile ./dirrpt/rcust_s.dsc, Purge
Map demo.*, target demo.*;
```

Run the following command on GGSCI prompt

```
--Use the ADD REPLICAT command to create a Replicat group
$GGSCI > ADD REPLICAT RCUST_S, INTEGRATED, EXTTRAIL ./dirdat/es
```

Start all the Extract and Replicat processes on the source and target instances and verify that there's no lag for the output column Lag at chkpt and Time since chkpt. Use the following GGSCI commands to start the processes on source.

```
$ ./ggsci
$ GGSCI (ip-source) > info all
$ GGSCI (ip-source) > START EXTRACT ECUST_S
$ GGSCI (ip-source) > START EXTRACT PCUST_S
$ GGSCI (ip-source) > STRAT REPLICAT RCUST_T
```

The following screenshot refers to the GoldenGate processes (Manager, Extract, Pump, and Replicat) running from the source RDS Custom for Oracle instance.

```
GGSCI (ip-10.0.10.10) 1> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	ECUST_S	00:00:00	00:00:08
EXTRACT	RUNNING	PCUST_S	00:00:00	00:00:00
REPLICAT	RUNNING	RCUST_T	00:00:00	00:00:07

Use the following GGSCI commands to start the processes on target.

```
$ ./ggsci
$ GGSCI (ip-target) > info all
$ GGSCI (ip-target) > START EXTRACT ECUST_T
$ GGSCI (ip-target) > START EXTRACT PCUST_T
$ GGSCI (ip-target) > STRAT REPLICAT RCUST_S
```

The following screenshot refers to the GoldenGate processes running from the target instance.

```
GGSCI (ip-10.0.10.10) 1> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	ECUST_T	00:00:00	00:00:06
EXTRACT	RUNNING	PCUST_T	00:00:00	00:00:07
REPLICAT	RUNNING	RCUST_S	00:00:00	00:00:06

Perform bi-directional replication validation between the source and target with the following code:

Perform sample data insertion in source and target database using following SQL*Plus commands:

Insert data on Source:

```
SQL>ALTER SESSION SET nls_date_format='yyyy-mm-dd hh24:mi:ss';
SQL>insert into demo.heartbeat(ID,HEARTBEAT_DATE) values (1,sysdate);
SQL>insert into demo.heartbeat(ID,HEARTBEAT_DATE) values (3,sysdate);
SQL>insert into demo.heartbeat(ID,HEARTBEAT_DATE) values (5,sysdate);
SQL>insert into demo.heartbeat(ID,HEARTBEAT_DATE) values (7,sysdate);
SQL>commit;
SQL>select * from demo.heartbeat;
```

Insert data on target:

```
SQL>ALTER SESSION SET nls_date_format='yyyy-mm-dd hh24:mi:ss';
SQL>insert into demo.heartbeat(ID,HEARTBEAT_DATE) values (2,sysdate);
SQL>insert into demo.heartbeat(ID,HEARTBEAT_DATE) values (4,sysdate);
SQL>insert into demo.heartbeat(ID,HEARTBEAT_DATE) values (6,sysdate);
SQL>insert into demo.heartbeat(ID,HEARTBEAT_DATE) values (8,sysdate);
SQL>commit;
SQL>select * from demo.heartbeat;
```

Make sure the GoldenGate processes are running without any lag or errors. If the process is not **RUNNING**, verify the error by using GGSCI command **View Report <Process_name>**.

The following screenshot shows data validation from the source instance using SQL*Plus.

SourceDB

SQL> select * from demo.heartbeat;

ID	HEARTBEAT_DATE	AZ
1	2022-08-15 19:07:23	1a
3	2022-08-15 19:07:23	1a
5	2022-08-15 19:07:23	1a
7	2022-08-15 19:07:23	1a
2	2022-08-15 19:07:42	1b
4	2022-08-15 19:07:42	1b
6	2022-08-15 19:07:42	1b
8	2022-08-15 19:07:43	1b

8 rows selected.

The following screenshot shows data validation from the target instance using SQL*Plus.

targetdb

SQL> select * from demo.heartbeat;

ID	HEARTBEAT_DATE	AZ
1	2022-08-15 19:07:23	1a
3	2022-08-15 19:07:23	1a
5	2022-08-15 19:07:23	1a
7	2022-08-15 19:07:23	1a
2	2022-08-15 19:07:42	1b
4	2022-08-15 19:07:42	1b
6	2022-08-15 19:07:42	1b
8	2022-08-15 19:07:43	1b

8 rows selected.

To monitor GoldenGate processes for any replication lag, refer to the following GGSCI commands.

The following shows the replication lag monitoring on the source instance using GGSCI commands.

```
GGSCI (ip-██████████) 13> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	ECUST_S	00:00:00	00:00:02
EXTRACT	RUNNING	PCUST_S	00:00:00	00:00:00
REPLICAT	RUNNING	RCUST_T	00:00:00	00:00:07

```
GGSCI (ip-██████████) 14> dblogin USERIDALIAS rdscustom_source
Successfully logged into database.

GGSCI (ip-██████████) as gguser@RDSTEST) 15> LAG extract ECUST_S

Sending GETLAG request to EXTRACT ECUST_S ...
Last record lag 3 seconds.
At EOF, no more records to process

GGSCI (ip-██████████) as gguser@RDSTEST) 16> LAG extract PCUST_S

Sending GETLAG request to EXTRACT PCUST_S ...
No records yet processed.
At EOF, no more records to process

GGSCI (ip-██████████) as gguser@RDSTEST) 17> LAG REPLICAT RCUST_T

Sending GETLAG request to REPLICAT RCUST_T ...
No records yet processed.
Low watermark lag: 1.
High watermark lag: 1.
Low watermark position: 3995652.
High watermark position: 3995653.
At EOF, no more records to process
```

The following shows the replication lag monitoring on the target instance using GGSCI commands.

```
GGSCI (ip-██████████) 1> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	ECUST_T	00:00:00	00:00:07
EXTRACT	RUNNING	PCUST_T	00:00:00	00:00:03
REPLICAT	RUNNING	RCUST_S	00:00:00	00:00:07

```
GGSCI (ip-██████████) 2> dblogin UserIdAlias rdscustom_target
Successfully logged into database.

GGSCI (ip-██████████ as gguser@ORCL) 3> LAG EXTRACT ECUST_T

Sending GETLAG request to EXTRACT ECUST_T ...
Last record lag 3 seconds.
At EOF, no more records to process

GGSCI (ip-██████████ as gguser@ORCL) 4> LAG EXTRACT PCUST_T

Sending GETLAG request to EXTRACT PCUST_T ...
No records yet processed.
At EOF, no more records to process

GGSCI (ip-██████████ as gguser@ORCL) 5> LAG REPLICAT RCUST_S

Sending GETLAG request to REPLICAT RCUST_S ...
No records yet processed.
Low watermark lag: 1.
High watermark lag: 1.
Low watermark position: 5012736.
High watermark position: 5012737.
At EOF, no more records to process
```

What happens during failover

In a failover scenario, one of the RDS Custom for Oracle instances is unavailable. If we're using Route 53 or a proxy as a solution, it automatically directs all the application traffic to the RDS Custom instance that is available. Because this solution uses Oracle GoldenGate bi-directional replication, the RPO is 0.

At this point, Amazon RDS Custom for Oracle is running as a standalone database, without any HA or DR. After the application failover is complete and tested, we can focus on recreating the HA architecture.

The following are the high-level steps to recreate the HA architecture after failover:

1. If the RDS Custom source instance in AZ1 goes down, you can divert the application traffic to the RDS Custom target instance in AZ2. If you're using Route 53 or a proxy, then it is handled automatically.
2. Create the new RDS instance in AZ1 as a read replica from the running RDS Custom for Oracle instance in AZ2.
3. Set the name to that of the instance that failed .
4. At this point, the Oracle Golden Gate Capture process keeps capturing all the transactions to trail files from the RDS Custom instance running in AZ2, but the pump (Extract) process goes into the ABENDED state because the manager process is down currently not running because RDS Custom instance in AZ1 is not running.
5. Promote the read replica to a standalone RDS Custom for Oracle instance in AZ1.
6. The Amazon EFS that was mounted on Failed RDS Custom on AZ1 will still have all the trail files and GoldenGate shared files.
7. After the instance is successfully promoted, mount the same Amazon EFS file system that was mounted on the failed RDS Custom instance in AZ1
8. Reconfigure all the GoldenGate processes on RDS Custom instance in AZ1 and start the Extract (pump) process on the GoldenGate running on RDS Custom instance in AZ2.
9. After all the GoldenGate processes are started successfully on RDS Custom instance in AZ1 and AZ2, GoldenGate starts receiving all the backlog Replicat trail files on the RDS Custom instance in AZ1.
10. Replicat makes checkpoints in the trail file to mark its last read position.
11. When the replication lag is zero, all the pending transactions are applied to AZ1 to make sure the source and target are in sync.

What to expect during database instance recovery

You could experience scenarios like database point-in-time recovery, recovery from failed instances, read replica promotions, and so on, when you may have to perform database restore and recovery steps on the RDS Custom instance. Use the following high level checklist to validate the configuration after the recovery.

1. After the database restore has been completed, you must reconfigure and validate database-related parameters before restarting the Oracle Golden Gate processes. For example, `ENABLE_GOLDENGATE_REPLICATION` must be `TRUE` .
2. Validate and verify the Amazon EFS file system is mounted to the restored instance. Refer to [Integrate Amazon RDS Custom for Oracle with Amazon EFS](#) for more information.
3. Validate the configuration of the automatic remount of the Amazon EFS file system on reboot of the instance under `/etc/fstab` .
4. Validate the symbolic links from the Oracle GoldenGate home directory to the Amazon EFS file system have been reestablished after restore.
5. Validate the Oracle GoldenGate binaries are accessible on new instance.
6. Update the Oracle Home TNS entries with the updated host endpoint on both the source and target DB instances under `$ORACLE_HOME/network/admin/tnsnames.ora` .

7. Invoke `GGSCI>` on the newly restored instance and validate the `CREDENTIALSTORE` for the successful connection by making `dblogin USERIDALIAS`.
8. Start the GoldenGate Manager process on the newly restored RDS Custom instances.
9. Start the GoldenGate `ABENDED/STOPPED` Extract process on newly restored RDS Custom instances. Whenever we start the `ABENDED` Extract processes, the process is getting recovered from the last `ABEND` point. The Extract process searches back to its online redo log file or archive log file record for the opened transactions when it crashed. Make sure the archive files are available for Extract process recovery to avoid any data loss. If needed, [register](#) the Extract to return to an existing LogMiner data dictionary build with a specified SCN. This allows for faster creation captures by using existing dictionary builds. See [Restarting a Primary Extract after System Failure or Corruption](#) for more details.
10. Start the GoldenGate extract Pump process by updating the rmthost Pump process parameter with the updated host entry. The extract Pump process will `ABEND` with the following error if the host address isn't updated in the GoldenGate parameter files: `TCP/IP error 111 (Connection refused), endpoint`.
11. Start the Replicat process and Oracle GoldenGate will pick up the trail file information from the checkpoint table. See [Starting Replicat after Point In Time Restore Of Database](#) for more information.
12. In case of any error, follow the Oracle GoldenGate [troubleshooting](#) best practices.

Conclusion

In this post, we showed you how to configure and use Oracle GoldenGate bi-directional replication with Amazon RDS Custom for Oracle to achieve near zero HA and/or DR with near zero data loss and minimal downtime using Amazon EFS.

In [Part 2](#) of this series, we discuss how to make this architecture even more resilient and provide HA and DR with zero Data loss in the event of a AZ failure and near zero data loss in the event of a Region failure.

To learn more about Amazon RDS Custom and Amazon EFS, refer to the following resources:

- [Amazon RDS Custom pricing](#)
 - [Amazon RDS Custom User Guide](#)
 - [Amazon Elastic File System Documentation](#)
 - [Implement Oracle GoldenGate high availability in the AWS Cloud](#)
-

About the authors



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